Small Model, Honest Numbers: A Vocabulary-Pruned DeBERTa-v3-xsmall for RTE

[Student name] [Roll number] CS6315E Neural Networks and Deep Learning
Code: `https://github.com/ihemanthm/rte-challenge`

1 Approach

Model choice. Since accuracy is weighed against size, I compared three sizes of one pretrained family, DeBERTa-v3 [2, 3] (base 184M, small 142M, xsmall 71M), with identical code. v3-base is my reference baseline and the distillation teacher; v3-xsmall is the student.

Fine-tuning protocol. One config-driven training path for all models: AdamW, 10% linear warm-up then decay, layer-wise LR decay 0.9, separate head LR, max length 256, batch 16, early stopping on validation accuracy. With 2,490 training pairs results swing with the seed [7], so every configuration uses 5 seeds (mean ± sd). All decisions (early stopping, checkpoint and model selection, pruning level) use a 10% validation split of train that is *grouped by premise*, so near-duplicate pairs never straddle it. The dev set (277) is used only to report.

Distillation. Teacher soft labels are computed *out-of-fold*: five DeBERTa-v3-base models, each trained on four premise-grouped folds, predict the fifth (OOF accuracy 82.9%). The student minimises $\alpha\,\mathrm{CE}(y, s) + (1-\alpha)\,T^2\,\mathrm{KL}(\sigma(t/T)\,\|\,\sigma(s/T))$ [4] with α=0.5, T=2, plus two variants: *masked* KD (pure CE on rows the teacher gets wrong) and α=0.8, T=4.

Vocabulary pruning. 49.2M of xsmall's 70.8M parameters are the 128,100 × 384 word embedding, yet RTE uses about 19k tokens. After training, I keep only the embedding rows of: special tokens ∪ every training-set token ∪ the top-N tokens ranked by the tokenizer's own unigram log-probabilities [5]; all other ids map to `[UNK]` through a fixed lookup table [1]. On input made only of kept tokens the pruned model is *exactly* the original. N is chosen on train alone: the smallest N for which one half of train covers 98% of the other half's tokens (giving N=60,000, 62,245 rows kept). I also tried a rank-128 SVD factorisation of the embedding (38.1M parameters).

2 Results

The submission has **4.0× fewer parameters** than DeBERTa-v3-base (45.5M vs. 184.4M) and 2.1× fewer non-embedding parameters than v3-small. Against my v3-base baseline it is 5.4 points lower on dev for the selected checkpoints (85.6 vs. 80.1; McNemar p=0.058, 35 vs. 20 discordant pairs) and about 7 points lower on 5-seed means. Pruning flipped 10 of 277 dev predictions (6 wrong, 4 right; p=0.75).

Distillation did not improve accuracy. KD at α=0.5 scored −0.9 points versus plain fine-tuning (paired $t = -1.29$ over seeds; McNemar p=0.31: not significant). It did make the student better calibrated (dev log-loss 0.74 vs. 0.99) and more stable across seeds (92% vs. 88% pairwise prediction agreement). The reason is a bias in the teacher: on train it predicts *entailment* for 57.8% of pairs against 50.2% gold, and 72% of its OOF errors are on not-entailment pairs. The distilled student inherits it (61.5% entailment predictions on dev). The masked and softer variants test whether removing the wrong soft labels fixes this.

Model	Total params	Non-emb.	Dev acc. (5 seeds)	Val acc.
Majority / word overlap / TF-IDF + LR	–	–	52.7 / 57.4 / 57.0	–
Instructor reference baseline	[TBD]	[TBD]	[TBD]	–
DeBERTa-v3-base (my baseline, teacher)	184.4M	86.0M	84.8 ± 1.1	86.2
DeBERTa-v3-small	141.9M	43.5M	76.5 ± 1.4	75.1
DeBERTa-v3-xsmall, plain fine-tuning	70.8M	21.6M	77.7 ± 2.1	78.5
+ KD (α 0.5, T 2)	70.8M	21.6M	76.8 ± 1.6	78.4
+ masked KD (α 0.5, T 2)	70.8M	21.6M	[pending]	[pending]
+ KD (α 0.8, T 4)	70.8M	21.6M	[pending]	[pending]
rank-128 factorised embedding	38.1M	21.6M	51.0 ± 5.0	57.6
Submission: best-validation xsmall checkpoint, vocab-pruned	**45.5M**	21.6M	**80.1** (1 ckpt)	81.0
same checkpoint before pruning	70.8M	21.6M	80.9 (1 ckpt)	81.0

Table 1: GLUE RTE dev accuracy (%). "Val" (mean best validation accuracy) is the only number used for selection. The submission is one checkpoint (seed 21) picked by validation, so its dev score is optimistic; the 5-seed mean (77.7) estimates the method.

3 Accuracy vs. parameters

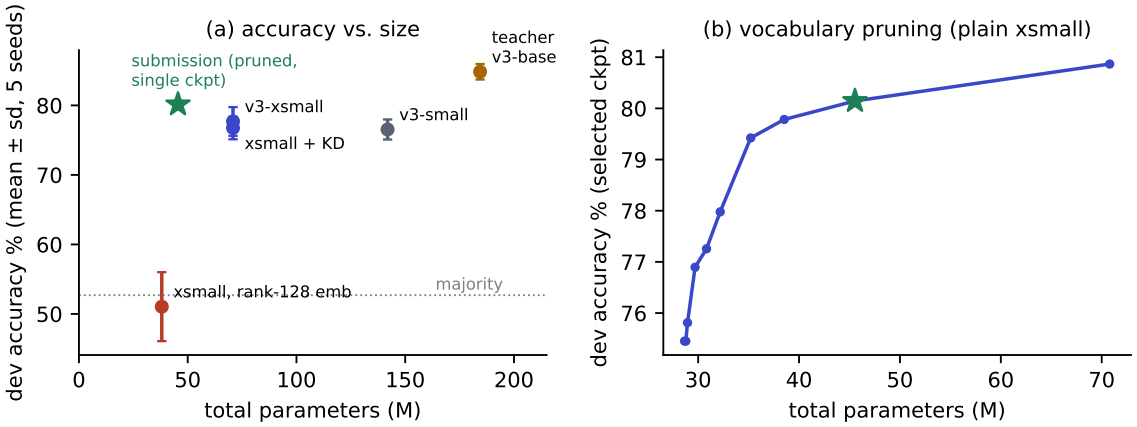


Figure 1: (a) Dev accuracy (mean ± sd over 5 seeds) against total parameters; the star is the pruned submission (single checkpoint). (b) Pruning curve for the selected xsmall checkpoint: train tokens plus the top-N frequent tokens, $N \in \{0, \ldots, 60\text{k}, \text{all}\}$.

Larger is not automatically better here (Fig. 1a). v3-small has twice xsmall's parameters and twice its non-embedding parameters, but scored lower on both validation (75.1 vs. 78.5) and dev (76.5 vs. 77.7); it fits the 2.2k training pairs within 3–4 epochs and then only memorises. The large jump is to v3-base (+7 points), which is wider (768 vs. 384 hidden) *and* deeper than small, so on this data the benefit comes from pretraining capacity per layer rather than from raw parameter count, most of which is embedding table in all three.

Where parameters can be cut matters more than how many. Shrinking the embedding by SVD at rank 128 kept only 57.5% of its energy; training loss stayed at chance (0.70) for 4–5 epochs on every seed and the model never recovered (51.0%). Pruning the vocabulary instead removes rows that never fire, so the remaining model is unchanged: −36% parameters for −0.7 points. Fig. 1b shows a smaller version would do worse: keeping only training tokens (28.7M) costs

5.4 points, because unseen dev tokens are mostly rare names, and only 98.5% of dev tokens (60% of pairs) are fully covered even at N=60k.

4 Error analysis

The submission makes 55 errors on dev. 34 are false *entailment* predictions (error rate 26.0% on gold not-entailment vs. 14.4% on entailment), so the model over-predicts entailment (57.4% vs. 52.7% gold). Automatic tags, error rate on each subset (overall 19.9%): high lexical overlap ($\geq$80%) but not entailed 42.1% (n=38); low overlap (<50%) but entailed 42.9% (14); top-10% long premises 28.6% (28); numbers/dates in the hypothesis 27.1% (48); negation in exactly one side 25.0% (40), so negation is not a major error source here.

I hand-labelled a random sample of 25 errors (seed 0; `report/error_annotation.csv`):

- **Overlap, unsupported claim (6, all false entailments).** The hypothesis reuses premise words but changes a role or adds a detail: "Doha, the capital of Qatar" → "Qatar is located in Doha"; "a substance found in red wine" → "red wine". This is the lexical-overlap heuristic known from NLI models [6].
- **Paraphrase missed (5, false non-entailments).** "BSE, the so-called mad cow disease", "NYT column" → works for the *New York Times*; two of these sit inside long premises.
- **World knowledge (5).** Four entailments need background facts (Renault is European; asked to form a government ⇒ new prime minister). In one false entailment the model used outside knowledge the premise does not state (George W. Bush's wife is Laura).
- **Modality, quantifiers, precision (4).** "asked the government to investigate" → "investigates"; "ready to be used" → "is used"; "only those who can afford" → "all parents"; "nearly 4 percent" → "4 percent".
- **Other (5):** number binding (32 *populations* read as 32 pandas; an age attached to the wrong person), two near the threshold ($p \approx 0.42$), one doubtful gold label.

5 Reflection

Counting *where* the parameters are paid off most: post-hoc vocabulary pruning is cheap, exact on covered text, and gave the largest efficiency gain. Logit distillation from a biased teacher, factorising the embedding and a bigger student did not help. **With more time/compute:** (i) prune the vocabulary *before* fine-tuning and use the saved budget for a wider backbone; (ii) intermediate-layer or attention distillation instead of logits only, with the teacher's entailment bias corrected (class re-weighting or temperature per class); (iii) train-only data augmentation targeting the error classes above (role swaps, modality and quantifier edits of training hypotheses as not-entailment); (iv) a 6-layer xsmall to extend the size curve downward; (v) more seeds to tighten the ±2-point seed spread, which is larger than most effects measured here.

Reproducibility. Pinned `uv` environment; every run stores config, package versions, parameter counts and per-seed results; two teacher runs gave byte-identical OOF logits. `make venv data train submission` reproduces the submission.

References

[1] A. Abdaoui, C. Pradel, G. Sigel. Load What You Need: Smaller Versions of Multilingual BERT. *SustaiNLP*, 2020.

[2] P. He, X. Liu, J. Gao, W. Chen. DeBERTa: Decoding-enhanced BERT with Disentangled Attention. *ICLR*, 2021.

[3] P. He, J. Gao, W. Chen. DeBERTaV3: Improving DeBERTa using ELECTRA-Style Pre-Training with Gradient-Disentangled Embedding Sharing. *ICLR*, 2023.

[4] G. Hinton, O. Vinyals, J. Dean. Distilling the Knowledge in a Neural Network. *arXiv:1503.02531*, 2015.

[5] T. Kudo. Subword Regularization: Improving Neural Network Translation Models with Multiple Subword Candidates. *ACL*, 2018.

[6] R. T. McCoy, E. Pavlick, T. Linzen. Right for the Wrong Reasons: Diagnosing Syntactic Heuristics in Natural Language Inference. *ACL*, 2019.

[7] M. Mosbach, M. Andriushchenko, D. Klakow. On the Stability of Fine-tuning BERT. *ICLR*, 2021.